

Prottus Manna

+91-8697736679 | prottus2004@gmail.com | [linkedin.com/in/prottus-manna](https://www.linkedin.com/in/prottus-manna) | github.com/prottus2004 | prottus-visual-showcase.vercel.app | Kolkata, India | Open to Relocation

EDUCAT

Institute of Engineering and Management (IEM), Kolkata – MAKAUT

2022 – 2026

Bachelor of Technology in Information Technology

SGPA: 9.00 in Final Semester

National Gems Higher Secondary School, Behala

2020

Class X – ICSE (Indian Certificate of Secondary Education)

Aggregate: 86%

National Gems Higher Secondary School, Behala

2022

Class XII – ISC (Indian School Certificate)

Aggregate: 76%

Work Experience

Fraud Guard AI | *Python, React.js, Node.js, Express.js, REST API*

June 2025 – July 2025

- Designed and deployed a full-stack production application with REST API integrations to third-party email/data services, structuring the backend for modular, independently deployable services.
- Built an **automated** retraining pipeline triggered by analyst feedback, enabling continuous deployment of model updates with zero manual intervention – improving detection accuracy by **~18%** and system **reliability**.
- Applied Agile/Scrum practices and **CI/CD**-oriented delivery across the full SDLC in **collaboration** with a cross-functional team; authored technical design docs and API specs supporting **scalability**, maintainability, and long-term **operational excellence**.

Vision AI Suite | *Python, PyTorch, CNN+LSTM, U-Net, OpenCV, Streamlit*

May 2025 – June 2025

- Architected and deployed a full-stack concurrent multi-model deep learning pipeline running three specialised neural networks simultaneously: a CNN+LSTM encoder-decoder for image captioning, a U-Net for semantic segmentation, and a custom object detection module.
- Deployed the complete AI application pipeline as a real-time interactive Streamlit app, supporting live video and image inputs with optimised low-latency concurrent execution across all three model architectures.
- Created an efficient multi-model serving strategy to minimise GPU memory usage and reduce latency – demonstrating the ability to triage and resolve system performance issues at the hardware and service layer.

Key Projects

Optimus AI — Agentic Voice Assistant

July 2025 – Present

Python, LangChain, LangGraph, LLMs (NIM/Ollama), RAG, edge-TTS, SQLite, Selenium, PyAutoGUI, Pydantic

- Architected an end-to-end **agentic AI pipeline** mapping a single spoken command to multi-step action: LLM intent router → Pydantic-typed tool registry (20+ tools) → supervised **self-healing executor** (LLM-guided retry on failure) → natural-language response generator, with wake-word detection and voice barge-in.
- Built a **pluggable multi-provider LLM chain** (NVIDIA NIM → local Ollama) behind a uniform interface with automatic health-based fallback, and a **date-aware RAG pipeline** (LLM query reformulation, recency filters, multi-source web grounding) for hallucination-free real-time answers on live facts, prices, and news.
- Engineered a low-latency **streaming TTS engine** (4-worker parallel synthesis + sentence chunking + instant “thinking” filler) and **runtime OS-level app discovery** fusing 5 sources (Start Menu shortcuts, registry App Paths, Uninstall entries, PATH, web fallback) with fuzzy reconciliation – zero hardcoded app lists, portable across Windows machines.
- Implemented a **disk-first code-generation agent** (plan → 70B-model streaming generation → atomic save → IDE focus-guarded typing → syntax verification with LLM-assisted regeneration) and **SQLite-backed memory** for conversation continuity and preference learning across sessions.

Achievements & Recognition

Best Intern Award – Zidio Developments (Aug 2025): Scored 83% on final evaluation on production-grade application engineering projects as the sole engineer on the team.

NUS Study Visit – AI & ML: Selected by IEM Kolkata to visit National University of Singapore; received specialised training in Artificial Intelligence, Machine Learning, IoT, and Data Analytics at the National University of Singapore.

Coding Efficiencies with real-time projects : Active Codechef and Leet Code Problem solver

Technical Skills

Programming Languages: Python, Java, C/C++

Full-Stack Development: React.js, Node.js, Express.js, REST API Design, JSON, HTTP/HTTPS, Streamlit

AI / ML & LLMs: PyTorch, TensorFlow, scikit-learn, Large Language Models (LLMs), RAG Pipelines, Prompt Engineering, NLP, CNN, LSTM, U-Net, Transformer Models, Supervised/Unsupervised Learning

System & Application Integration: REST API, Third-Party API Integration, Sync Integration, Automated Retraining Pipelines

Databases: MongoDB (NoSQL), MySQL, PostgreSQL, SQL Query Optimisation

Cloud & DevOps: Microsoft Azure (Fundamentals), Git, GitHub, Linux, CI/CD Pipelines, Agile/Scrum

Core CS Concepts: Data Structures & Algorithms (DSA), OOP, System Design, SDLC, Operating Systems, DBMS, Computer Networks

Engineering Practices: Clean/Modular Code, Technical Documentation, Issue Triage & Debugging, Test Documentation, Code Review

Full-Stack Web Development: HTML, CSS, JavaScript, Frontend/Backend, Hosting, Monitoring & Logging, Performance Optimization, Security, Testing, DevOps & Deployment, API Integration & Communication, Version Control & Collaboration